

MGMT 59000 Visual Analytics

Team Project: Executive Pitch Assignment

Project Overview

Deliverable: One killer visualization + pitch presentation + 3-page written report

Context

You are pitching a business case to a group of senior technology executives at your company. However, a critical call came in, and the **CTO will have to leave the meeting in 5 minutes**. This is your only opportunity to secure funding for your project.

The Challenge

You have **exactly 5 minutes** to summarize your key findings, assumptions, and/or projections to convince your CTO that **YOUR project** is worth funding.

Key Constraint: Your presentation must be compelling enough to secure buy-in before the CTO leaves the room.

Project Objectives

This assignment is designed to develop three critical skills:

1. **Crafting a compelling story with data** - Learn to identify meaningful patterns and insights that drive business decisions
2. **Clearly articulate that story with ONE killer figure** - Master the art of distilling complex analysis into a single, powerful visualization
3. **Be creative and have fun** - Explore innovative approaches to data visualization and storytelling

Your Tasks

1. Identify Your Domain

Select a domain that is interesting and relevant to your team. Consider areas such as:

- Healthcare analytics
- Financial forecasting
- Supply chain optimization
- Customer behavior analysis
- Climate and environmental data
- Social media trends
- Transportation and logistics
- Education and learning analytics

Tip: Choose a domain where data-driven insights can drive clear business value.

2. Find a Dataset

Locate a high-quality dataset in your chosen domain. Common resources include:

- **HuggingFace Datasets** (<https://huggingface.co/datasets>)
 - **Kaggle** (<https://www.kaggle.com/datasets>)
 - **UCI Machine Learning Repository** (<https://archive.ics.uci.edu/ml/index.php>)
 - **Papers with Code** (<https://paperswithcode.com/datasets>)
 - **Data.gov** (for government datasets)
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3. Craft Your Project Idea

Develop a compelling project concept based on your dataset.

Crafting Your Idea: Key Questions

Before diving into analysis, answer these fundamental questions:

1. What problems are you trying to solve?

- Identify the specific business challenge or pain point
- Understand the current state and desired future state
- Quantify the impact of the problem (cost, time, quality, etc.)
- Consider who is affected and how

2. What questions are you trying to answer?

- Define specific, answerable questions that address the problem
- Prioritize questions that lead to actionable insights
- Ensure questions align with executive decision-making needs
- Examples:
 - "Which customer segments are most likely to churn?"
 - "What factors drive the highest ROI in our campaigns?"
 - "Where are the bottlenecks in our supply chain?"

3. What kind of data will you need?

- Identify required data sources and variables
- Assess data availability and quality
- Consider temporal aspects (historical trends, seasonality)
- Think about granularity (daily, weekly, by region, by product)
- Determine if additional data enrichment is needed

Your Project Should:

- Address a clear business problem or opportunity
 - Provide actionable insights that drive decisions
 - Demonstrate potential ROI or strategic value
 - Be realistic in scope for implementation
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4. Create One Killer Figure

Design a single, powerful visualization that tells your entire story. This is the centerpiece of your pitch.

Your figure should:

- ✓ Clearly communicate your key findings or insight
- ✓ Be immediately understandable (within 30 seconds)
- ✓ Support your business case with data
- ✓ Be visually compelling and professional
- ✓ Highlight patterns, trends, or anomalies that matter
- ✓ Show projections or scenarios where relevant

Avoid:

- ✗ Overly complex visualizations that require extensive explanation
- ✗ Multiple disconnected charts (use one integrated figure)
- ✗ Poor color choices or unclear labeling
- ✗ Cluttered layouts with excessive information

Just remember all the things we've talked about during our "show and tell".

5. 5-Minute In-Class Presentation

Deliver your pitch using your killer figure as the visual anchor.

Presentation Format & Timing

Total Time: 15 Minutes Per Team

Each team's presentation will be structured as follows:

5 Minutes: Context Setting

- Introduce your domain and business problem
- Explain your dataset and methodology
- Set up the story you're about to tell
- Establish why this matters to the business

5 Minutes: Killer Figure & Pitch

- Present your ONE killer visualization
- Walk through key insights and findings
- Make your business case with data
- Deliver your funding request and expected impact
- **This is the critical moment where you convince the CTO**

5 Minutes: Q&A

- Field questions from your classmates (acting as the CTO and executives)
- Defend your assumptions and projections
- Clarify your methodology if needed
- Address concerns about implementation

Class Participation

Role-Playing: The rest of the class will act as the CTO and senior technology executives, asking tough questions about:

- Business viability
- Data quality and assumptions
- Implementation challenges
- ROI projections
- Alternative solutions

Voting & Bonus Points

Class Voting Process

After all presentations are complete, the class will vote on which project they would fund:

- ✓ One vote per person
- ✓ You CANNOT vote for your own team
- ✓ Vote for the project with the most compelling business case and presentation

Bonus Points Opportunities

Most Votes Bonus: +5%

The team that receives the most votes will earn a **5% bonus** on their project grade.

This recognizes excellence in:

- Compelling storytelling
- Effective visualization
- Strong business case
- Presentation quality
- Q&A performance

Creative Team Outfit Bonus: +3%

Teams that show up in **creative, fun, and coordinated outfits** will earn a **3% bonus** on their project grade.

Examples:

- Matching bow ties
- Themed costumes related to your project
- Color-coordinated professional attire
- Creative accessories that tie to your domain

Note: Outfits should be workplace-appropriate (don't show up in your Speedo, even if it matches the theme or your project)!

Evaluation Criteria

Your project will be assessed on:

Presentation Components (60%)

1. Business Relevance & Problem Definition (15%)

- Is this a real, meaningful business problem?
- Is the business case clearly articulated?
- Does the project align with strategic priorities?

2. Data Quality & Appropriateness (10%)

- Is the dataset suitable for the problem?
- Is data preprocessing appropriate?
- Are data limitations acknowledged?

3. Killer Figure Effectiveness (20%)

- Does the visualization clearly communicate the story?
- Is it immediately understandable?
- Are design choices appropriate and intentional?
- Does it support the business case?

4. Presentation Quality & Delivery (10%)

- Is the 5-minute pitch clear and compelling?
- Does the team manage time effectively?
- Is the narrative well-structured?
- Is the delivery professional and engaging?

5. Q&A Performance (5%)

- Does the team handle questions confidently?
- Are responses clear and well-reasoned?
- Does the team defend assumptions appropriately?

Written Report (40%)

6. Context & Problem Statement (8%)

- Clear articulation of the business problem
- Well-defined scope and objectives
- Compelling rationale for why this matters

7. Data Documentation (8%)

- Thorough description of data sources
- Clear explanation of data characteristics
- Appropriate discussion of limitations and preprocessing

8. Design Process & Rationale (12%)

- Thoughtful explanation of inclusion/exclusion decisions
- Documentation of iterations and evolution
- Clear justification for visual design choices
- Evidence of refinement based on feedback

9. Challenge Reflection (8%)

- Honest identification of a significant challenge
- Thoughtful analysis of how it was addressed
- Meaningful learning takeaways

10. Writing Quality & Formatting (4%)

- Clear, professional writing
- Proper formatting and organization
- Adherence to 3-page limit
- Proper citations

Bonus Points

Most Class Votes: +5% to final project grade

Creative Team Outfit: +3% to final project grade

Total Possible Score: 100% + up to 8% bonus = 108%

Final Submission Requirements

1. Presentation Day Deliverables

- Killer figure (high-resolution image or interactive link)
- Presentation slides (if used beyond the main figure)
- 15-minute presentation (5 min context + 5 min pitch + 5 min Q&A)

2. Written Report (3 Pages Maximum)

Submit a concise written report that includes the following sections:

Section 1: Context & Problem Statement

- Describe the business domain and problem you're addressing
- Explain why this problem matters and who it impacts
- Articulate the key questions you're trying to answer
- Define the scope of your project

Section 2: Data Overview

- Identify your data source(s) and provide citations
- Describe the dataset characteristics:
 - Size (number of records, variables)
 - Time period covered
 - Key variables and their meanings
 - Data quality observations
- Explain any data preprocessing or cleaning performed
- Discuss any limitations or biases in the data

Section 3: Killer Figure Design Process

Document your visualization development journey:

Design Decisions:

- What did you choose to include in the figure, and why?
- What did you deliberately exclude, and why?
- How does the figure support your business narrative?
- What visualization type(s) did you select and why?

Iterations & Evolution:

- Describe early versions or alternative approaches you explored
- What feedback did you receive (from teammates, test audiences)?
- How did the figure evolve through iterations?
- Include sketches or screenshots of earlier versions if helpful

Section 4: Biggest Challenge

Reflect on the most significant challenge your team encountered:

- What was the challenge? (technical, conceptual, team dynamics, data-related, etc.)
 - Why was it difficult?
 - How did you attempt to address it?
 - What was the outcome?
 - What did you learn from this challenge?
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Due Dates:

- **Presentation Date:** October 6th and October 8th in class
 - **Written Report Due:** October 10th, 11:59 pm
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Remember: You have 5 minutes to change the CTO's mind. Make every second count.